The 4E Approach of Cognition into User Evaluation Method in NIME

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ABSTRACT
The topic of evaluation has been extensively discussed and debated in the NIME community. Attempts have been made to explore what the word "evaluation" means for the NIME community, with findings suggesting that there are divergent understandings of the term. A review of papers published in the NIME conference proceedings revealed that the word is often used to denote the process of collecting feedback to improve a prototype. The term evaluation is also used to compare the devices and distinguish the specific tasks for what the DMI was developed.

In this paper, we provide an overview of the emerging paradigm of post-cognitivism and the aligned movement of enactivism which has roots in phenomenology and embodied cognition. We argue that the 4E approach of cognition can be used as a flexible resource by qualitative research exploring the enactive concept of sense-making perspectives. The idea of "embodiment" has come a central theme for NIME designers, since musical interactions involve creative and affective aspects. The 4E approach in musical experience is presented by a questionnaire. Task-based methods may be suited to examine usability, but the creative and embodied aspects experience within the NIME interaction is subjective and requires alternative approaches for evaluation. In this paper we propose evaluating two digital musical instruments beyond their technical qualities by integrating the 4E approach into user evaluation. We contextualize this discussion by presenting a questionnaire to the DMIs designer focused in the 4E approach of cognition.

Author Keywords
4E Approach Evaluation, Methods, User Study

1. INTRODUCTION
The evaluation represents a significant role in the process of designing digital musical instruments (DMI) [1]. In recent years, evaluation methods have been largely explored in NIME community [17]. There have been presented and discussed some methodologies for studying and evaluating interactions with digital musical instruments in NIME community [18, 20].

In this paper, we study evaluating digital musical instruments beyond their technical qualities by applying a subjective qualitative questionary integrated into the user experience within the 4E approach of cognition. To develop the method of evaluation we invited a DMI designer to participate on the questionary directly related to her own two devices.

We suggest the 4E approach of cognition as a practice to gather descriptions of the performance lived experience. We reference Varela's enactivism [3] as an account of the 4E approach. In line with the research on cognition and self-perception, our work moves from the acknowledgement of our subjective experience unfold through a questionnaire. We outline the 4E approach by interviewing process presenting how the method reveals the details of the DMI creator sensorimotor experience while performing.

In this work we suggest a useful way of complementing NIME's already impressive achievements on evaluating methods and of strengthening the kind of contribution that practice-led researchers can make. We intend to show the benefit of the enactivism discipline in NIME research in an endeavor to explore the sensorimotor sound activities while performing on DMIs.

In this exploratory research, we intend to give insights into how the term evaluation can be applied within the 4E approach of cognition looking for: a) introduce how the 4E approach might offer useful ways of thinking about and describing the creative development of musical agents and musical cultures; b) present how the perceptual boundaries of performing depend on the musical worldviews and by the interaction with the environments they are embedded in, and how such views evolve over time through the dynamic interaction with a range of components while improvising; c) Raise a reflexive evaluation related to the subjective 4E approach for DMI designers for a better understand about their sensorimotor activity while performing in their DMIs and the sense of intimacy with their instrument.

2. RELATED WORK
The task of evaluating DMIs is in fact strongly linked to that of designing them, and knowledge gained in any side of the equation should complement the other [17].

Wanderley and Orio propose to use a set of musical tasks for usability evaluation and also a set of relevant features to be tested in usability evaluations of controllers used in the context of interactive music: learnability, explorability, , and timing controllability [25]. Regarding Creative Process, Cavdir [6] propose a user study evaluation method that affects participants' interaction, creativity, and experience, in the context of movement-based musical interaction. Related to the effectiveness of an instrument, O'Modhrain suggested that in addition to evaluating the DMI by performance, there are many other characteristics to consider.as instrument builders, component manufacturers, customers and other factors [26]. Young and Murphy [37] propose that the evaluation of DMIs should focus on functionality, usability, and user experience. Reimer and Wanderley [29] recently wrote about the evaluation of user experience for NIMEs. Their research suggest that UX-focused evaluations typically were exploratory and that they were limited to novice performers and propose to use the "Musicians Perception of the Experiential Quality of Musical Instruments Questionnaire (MPX-Q)" to compare UX for different instruments.

Reed et. al [28] presents an epistemological structure of lived experience through micro-phenomenology as a beneficial methodology for studying and evaluating interactions with digital musical instruments.

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3. THE 4E APPROACH AS AN EVALUATING METHOD

The 4E approach of cognition is suggested as a practice for gathering descriptions of lived experiences in the context of sensory motor skills using the DMIs. The 4E approach (*embodied, embedded, enactive and extended*) suggests that cognition involves the whole body, as well as the body's situation in the environment [3]. The research objective within the 4E approach of cognition, [33] is subjective experience, in which the subject vs. the object can be integrated into the lived experience.

Schiavio and van der Schyff [30] highlight that "*embodied, embedded, extended, and enactive* aspects of the 4Es approach offer useful dimensions through which teachers, students, and researchers can begin to explore and evaluate the processes involved in musical improvisation.

This work uses the theory of enactivism in the approach of the 4E approach of cognition to better understand how musical affordances through improvisation using DMIs can bring significant results. As such, a qualitative evaluation was formulated based on the 4E approach. We present a case study that was elaborated focused on the 4E approach as a method of evaluation in the NIME ecosystem. We interview one NIME designer who responds about two DMIs developed by her.

In developing the methodology for this study, we have mostly been interested in the integration into real sensorial perspective scenarios within the 4E approach, keeping in mind that over time, performing abilities and expressivity play an important role. We find it important that practically all qualitative evaluations of NIMEs are carried out by inviting test subjects to explore the instruments for a brief period. We believe that a lot can be derived from accomplishing these first impression evaluations of DMIs, and we recognize that longitudinal evaluations demand time and cost [3].

There are different methods within longitudinal evaluation. This also applies to the data collection. The goal here is to evaluate the creative and embodied aspects through the 4E approach of cognition for an accurate personification of the user's experience when using a DMI within his/her specific environment context.

4. METHOD

Traditionally, evaluation methods for NIMEs and DMIs have largely been based on frameworks used in the field of Human–Computer Interaction (HCI) [8].

Wanderley and Orio [25] discuss the application of such methodologies in the evaluation of input devices for musical expression and propose to

use a set of musical tasks for *usability evaluation*. Frid writes that for a musical instrument, such tasks could focus on the production of musical entities and include the generation of isolated tones, i.e. pitches at different frequencies and loudness levels; basic musical gestures like glissandi, trills, vibrato, and grace notes; and musical phrases, such as scales and arpeggios, as well as more complex contours with different speeds and articulations [8].

Wanderley and Orio also propose a set of relevant features to be tested in usability evaluations of

controllers used in the context of interactive music: *learnability, exportability, feature controllability, and timing controllability* [25].

It is also clear that the traditional evaluation methodologies coming from HCI tend to be unsuited to the even more subjective evaluation of DMIs [4]. Since DMI designs can be evaluated

from multiple perspectives, di↵erent techniques and approaches are required [8].

We created a questionnaire related to the NIME ecological context based on the 4E approach where the designer is questioned about the sensory motor connections when playing the DMI created by him/her. The questionnaire is divided within the 4E approach following its main context that they overlay.

The invited designer for the evaluation presented the answers about two NIMEs within the epistemological questions from the 4E approach perspective.

As the answers are related to the two DMIs, in the questionnaire the upper cases **E_K** is related to the DMI Electronic_Khipu, and the **K_Y** related to the DMI Kanchay_Yupana.

4.1 Electronic Khipu

The Khipu has three-dimensional encoding and is a tangible computer system that requires the whole body and is presented in Figure 1-. Over and above the sense of sight involved, both touch and hearing are also fundamental parts. The voice was important in the traditional Khipu since the information presented in the knots was completed with speech and mnemonic exercises. With this device, the mind computes, and the hands knot the information that remains stored at a tangible level [4]. With these characteristics, the DMI designer built the Electronic Khipu as an instrument in a MIDI controller for experimental sound in which the interaction is based on the weaving of different kinds of knots in real time. In this case, making an analogy with the traditional device, hearing is necessary to understand the sonic knot in live performance.

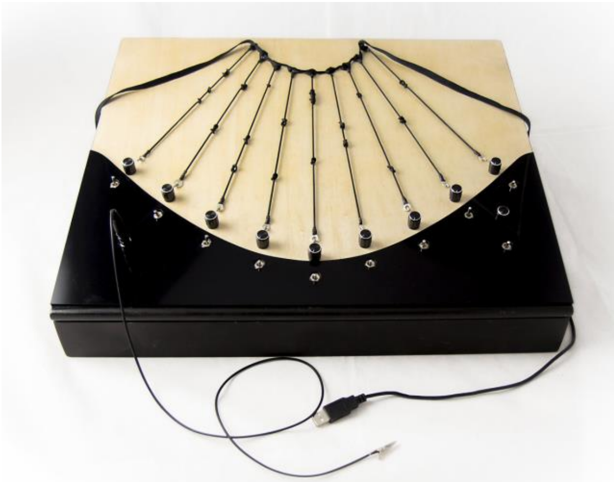

Figure 1. The Electronic Khipu

4.2 The Kanchay Yupana

This instrument was designed to complement the performance of the Electronic_Khipu. The yupana complemented the khipu in ancient times. The calculations resulting from the yupana were stored as encoded information in the knots of the khipus. In the premiere performance, as a ritual of gratitude, The Kanchay_Yupana// is played to mark the rhythm to complement the experimental sound produced by the manipulation of the Electronic Khipu by knotting its sensitive conductive rubber strings [5] as shown in Figure 2.

Figure 2. The Kanchay Yupana

5. THE 4E AND THE EVALUATION

5.1 Embodied

Considering the mind as embodied means reexamine the boundaries between the neural and extraneural factors (metabolic, thermodynamic, and muscular, among others) that drive cognitive processes. In this perspective, the brain becomes part of a larger network that involves the nervous system and the sensorimotor capacities of the entire organism [9]. In a sense, therefore, separating brain and body, perception and action, experience and behavior, may in fact be a largely artificial move that offers only limited accounts of what mental life involves [32]. Embodied cognition emphasizes the importance of the body in constraining intelligent behavior. The cognitive processes are not only computational processes carried out by the brain; rather, they should be explained by appealing to the physical features of the body that individuals of a species have [12].

5.1.1 Q &A

While performing using the DMI, do you have the sensation that your gestures and bodily actions are contributing effectively to your technical and sonic learning playing in that specific instrument [27]?

E_K: "*Absolutely! familiarising myself with the touch and the act of knotting each string totally contributes to the learning process. One of the characteristics of my instrument, in particular, is that it never sounds the same because many factors that do not depend on my technique influence it. However, it does help me a lot to know what kind of gestures are most effective".*

K_Y: "*It is different because by setting precise rhythms, the focus is directly on the previous preparation, which does not require specific gestures".*

If you become to perform using the DMI once more or subsequently, will your embodied memory allow you to recall and reapply your primary experience in new contexts, providing new perceptions and provoking melodic, harmonic and/or rhythmic intuition?

E_K: "*I have been using this instrument for about five years, and I have noticed that every time the interpretation is more intuitive, so I think that has to do with the embodied memory, it is worth mentioning that my instrument produces more abstract sounds than harmonics but this does not prevent it from being very expressive, and personally in my performance this is very connected with intuition and gives a lot of freedom to improvisation".*

K_Y: "*As it is a fully rhythmic interface, its continued use helps one learn the instrument and the type of patterns that work best for the piece being played. However, in my experience, this helps more in the composition process than in the performance process".*

How was your embodied memory knowledge different than semantic musical memories, which store general knowledge?

E_K & **K_Y: "***In my case, I only have a basic knowledge of music, so my musical semantic memory has been more influenced by the music I listen to and what I play in rehearsals. If it fits with my ideas (if it sounds good to me), colloquially, "having an ear", so in a way, although in principle embodied memory and musical semantic memory are different for me, they have a common thread".*

5.2 Embedded

Embedded minds are constituted by meaningful relationships that arise between agents and the environment. Embedded means that an embodied person is always in an environment and that meaning-making is shaped by the person's relationship and interactions with their physical and sociocultural environment [21].

5.2.1 Q &A

Regarding the NIME ecological context what interactive affordances the DMI provides within the environment?

E_K & K_Y: *Within the NIME environment, I consider that the most important possibility of both interfaces lies in the ability to learn about ancient instruments and technologies from territories made invisible in their adaptation to NIMEs. Thus, drawing a decolonial line makes new narratives and ways of creating, prototyping, and producing knowledge visible.* K_Y: *also offers significant didactic possibilities due to its easy learning. Thus, it also contributes to the teaching of basic musical semantics.*

In respect to your socio-cultural background, your living environment and your abilities, what are the musical cross-cultural affordances when performing by using the DMI?

E_K & K_Y: *Both interfaces have deep historical and cultural roots that enrich the performance when used with respect for their origins. My socio-cultural background is also vital in this context, as each performance becomes a personal ritual of contemplation and presence. For me, this is deeply connected to my spirituality and the way I honor the ancestral technologies that inspire the creation of my instruments. The way I engage with the audience is influenced by this perspective. I do not establish a hierarchy between performer and audience, Instead, I prefer to be at the same level as the audience, as is often seen in popular performances among many indigenous populations, allowing them to closely observe my interaction with the instruments and experience the sound in different ways. Although I have visuals that help the audience see the performance on a big screen, my instruments naturally spark curiosity. This curiosity helps me maintain the audience's attention, allowing many to perceive the ritual environment or, at the very least, engage in contemplation—even without the visuals.*

5.3 Enactive

The enactive dimension describes how organisms and their environments determine each other [33]. To better understand how this happens, we can consider the self-organized social dynamics of joint musical performance (e.g., a string quartet or a jazz trio) [34]. In such contexts, each performer is required to engage in circular processes of collaborative adaptive activity [3]; Here, individuality and collectivity must be continually renegotiated by each performer to sustain and develop the musical environment being enacted [36]. This involves multimodal online information from the individual parts being performed; an awareness of the music as it unfolds "as a whole"; the changing emotions and intentions of the musicians; their relationships to their instruments (and to their audience); as well as shared forms of visual, bodily, and auditory signaling, and more. Broader ecological factors, such as the nature of the acoustic space and the social significance of the musical event, are also important to consider. In short, musicians must individually and collectively adapt to a range of interactive dynamics in the local musical system [35]. They work within multiple levels of constraint to keep the musical environment coherent, maintaining their status as autonomous musical individuals whose actions simultaneously inform and are informed by the musical environment they co-promote. Furthermore, musical agents can and do push against such constraints to initiate transformations that keep the music vital or "alive" [30].

5.3.1 Q &A

Regarding your muscle memory, your bodily awareness, the sense of how your NIME feels in your hands or your intimacy with the instrument [19]; do these aspects enhance your capabilities in action musically, sonically, socially and emotionally while playing your NIME?

E_K Yes, one of the significant advantages of building an instrument from scratch is that you become familiar with every detail of its functionality, and your understanding of that instrument develops alongside it. In the case of the electronic Khipu, which is a non-traditional NIME the confidence in performance and expressiveness grows as you develop a body awareness that begins with the initial prototypes. Personally, the more I understand my interface—its sounds in response to touch, and even its reactivity to my mood due conductivity is a primary element—the richer the experience becomes, and the more confidence I gain in using it.

K_Y: In this case, the interaction is more straightforward, and its simplicity allows for quick integration into the instrument. As a result, the enactive experience may be less pronounced. However, I believe it still exists, as one gains confidence in using the NIME. Without this understanding, I would not be aware of its advantages and challenges.

5.4 Extended

While each of the Es in the 4Es approach offers a certain perspective on the nature of cognition, they are not disconnected. Rather, they overlap—aspects of one dimension will necessarily be reflected in the others. With this in mind, we conclude our analysis of the 4E approach with the "enlarged" aspect of cognition and creativity [7]. This is an important dimension to consider because, while many embodied approaches to cognition focus, by necessity, on the situated aspects of cognition, it is often argued that too much focus on neural and bodily factors can obscure the dynamic processes of codetermination that occur between (musical) agents and their environments (; Rowlands, 2010). The "extended" dimension of the 4E framework holds that "[biological and nonbiological] features of the environment can co-constitute the mind" [7]. Consider, for example, a percussionist improvising on an arrangement of instruments. In the process of enacting a meaningful relationship with this collection of instruments, the instruments must become part of the musician's cognitive ecology. In other words, the performer can be seen as "discharging" his or her musical expertise to the instruments in ways that are "functionally coupled" to the larger musical environment. This is to say that, when necessary, tools and objects from the environment can become integrative parts of mental life and the creative processes that accompany [34]. The very act of engaging with musical objects and technologies—for example, composing by improvising on a specific instrument, using music notation software (such as Finale or Sibelius), writing notes by hand on a sheet of paper, and so on—can contribute greatly to the development of musical ideas. In short, non-biological components of the physical world are important parts of the realm of musical cognition and creativity. And as technology develops, they may well become increasingly integrated into human bodies and brains [34].

5.4.1 Q &A

Does your NIME provide facilities to enhance your creativity by interacting with co-performers, technologies, and other non-organic ecological factors [19]?

E_K *Although this NIME is not designed for collaborative use, I have had the opportunity to perform alongside other artists using their own NIMES, which were inspired by similar principles, specifically the Andean Khipus. In these performances, the sounds coexisted to create an alternative soundscape, allowing me to explore aspects that I hadn't initially considered. This experience presented creative challenges that have certainly opened up new possibilities for this instrument.*

Collaborative performance with the Peruvian artist, Paola Torres Nuñez del Prado: https://www.youtube.com/watch?v=yA4hR9DS82k

K_Y: *This NIME was created out of the need for a rhythmic element to enhance the performance and guide the abstract sonority of the Electronic_Khipu. From its inception, it was designed to interact with other NIMEs and performances, serving consistently as a collaborative component, even with traditional instruments.*

Collaborative performance with artist Tita Anderson Lovell playing the quena, a traditional Andean music instrument: https://youtu.be/NPtHGYH0JV0?t=3595

Will the creative possibilities interacting with co-performers and other devices in different environmets enhance your technical and expressive knowledge enahncing your intimacy with your NIME [24]?

E_K & **K_Y:** *Collaboration offers valuable opportunities to understand NIMEs from different perspectives and to explore unique aspects that may not be discovered when working alone. While collaboration can present challenges and frustrations, it ultimately leads to both technical and performative experiences. This process not only deepens one's technical understanding of the NIME but also fosters stronger connections with the interface.*

6. DISCUSSION

7. CONCLUSION

We introduce the 4E approach to evaluate the experience of interaction within DMIs. We hope the use of the 4E approach in the study of implicit knowledge will benefit the NIME community as a new form of evaluation and research. This follows the community's growing interest in enactivist experiences [21] within DMIs [13, 14, 16] and expressivity and musicality while playing their instruments [11, 23]. Through the 4E approach evaluations carried out on our presented interview with two DMIs from the same designer, other experiential

dimensions were unfolded demonstrating a vast range of subtle details while performing with her own creations. We could observe how the 4E approach can be provide by specific formulated questions an impressive representation of the sensorimotor aspects of a designer performing with musicality and expressiveness her own instruments.
These suggestions of lived practice within the 4E approach evaluation through a questionnaire, could usefully contribute to considerations for designers in terms of the pursuit of embodied expressivity considering performances in evaluation.
We believe evaluating NIMEs within the 4Eapproach questionnaire presented in this work, can provide a more accurate analyzes concerning the perceptive capacities, musicality, the sense of intimacy with their instrument, expressiveness and affordances.

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